Smart Traffic Incident Detection & Emergency Dispatch

BACKGROUND

India records 1.54 lakh road deaths annually — the highest in the world. Around 70% of these fatalities are linked to delayed emergency response, with average response times ranging from 20–40 minutes in metro cities. Studies suggest that reducing emergency response time by just five minutes could save nearly 2,000 lives every year. Currently, most traffic incidents are identified manually through CCTV monitoring, resulting in delayed, inconsistent, and inefficient emergency handling. Existing systems also struggle with Indian mixed traffic conditions involving autos, two-wheelers, irregular lane discipline, and dense urban congestion.

PROBLEM STATEMENT

Design a real-time multimodal deep learning system capable of automatically detecting traffic incidents by fusing four heterogeneous data streams: CCTV video feeds (30 fps), roadside acoustic sensors (48 kHz), snapshot images captured every 5–10 seconds, and delayed text reports from emergency calls and IoT sensors. The system should classify incident severity (minor, major, fatal), localize the incident using GPS coordinates, predict the required emergency response units, optimize dispatch routing, and provide inference results with sub-500ms latency on edge hardware such as NVIDIA Jetson devices. The system must address major challenges including temporal misalignment across asynchronous data streams, extreme class imbalance, robustness to missing modalities, adaptation to Indian traffic conditions, real-time edge inference, explainability for operator trust, and integration with downstream emergency dispatch systems. The target performance includes detection F1-score ≥ 88%, severity classification accuracy ≥ 85%, false positive rate below 2 per hour, and ≥ 75% accuracy even when one modality is unavailable.